Overcoming Encryption Blindness: Zero-Day Anomaly Detection Using Graph and Transformer Neural Networks

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***Abstract*—Encryption protocols (TLS 1.3 and QUIC) have been widely adopted and have effectively deprecated traditional Deep Packet Inspection (DPI) and payload dependent machine learning methodologies used in network security. Additionally, current supervised learning architectures have serious limitations since zero-day vulnerabilities do not exist in historical training datasets and cannot be detected by static classification of neural networks. To solve these serious issues, we have propose a new hybrid deep learning architecture, GAT-E, which is designed to detect zero-day anomalies in highly encrypted network environments. Our proposed hybrid framework augments the model with a Self-Supervised Contrastive Learning (SSCL) methodology enabling the model to dynamically learn the mathematical boundaries of "normal" network behavior from massive amounts of unlabeled traffic. Our architecture generates a Flow Interaction Graph (FIG) to evaluate the entire network topology instead of only individual connections. The architecture uses a Sparse Transformer Encoder that extracts the temporal sequence and rhythm of packet flows and combines with a Graph Attention Network (GAT) to evaluate spatial topological features and neighborhood interactions. Initial experimental results with the CSE-CIC-IDS2018 and IoT-23 datasets indicate that GAT-E achieves a zero-day anomaly detection rate of 98.4% and greater than 4% higher than current state of the art supervised baselines while keeping the False Positive Rate (FPR) under 0.5% across fully encrypted environments.**

***Index Terms*—Network Intrusion Detection, Encrypted Traffic, Graph Attention Networks, Transformers, Self-Supervised Learning, Zero-Day Detection**

I. Introduction

Although 95% of internet traffic today is secured through Protocols like TLS 1.3 and QUIC, encryption has resulted in an "encryption blind spot" for traditional NIDSs [1], [2]. Legacy systems that use Deep Packet Inspection (DPI) and payload signatures are unable to detect malicious activity because they are unable to see the encrypted payload of packets [2], [3]. Traditional Machine Learning models for detecting attacks on networks typically treat network flows independently (like individual "rows" in a spreadsheet), and do not incorporate the interrelationship and coordinated behavior of advanced attacks [4]–[6].

The GAT-E architecture addresses these issues by using Flow Interaction Graphs (FIG) to represent network traffic instead of payloads and the communication topology [7], [8]. By using a "graph" representation, GAT-E can model the relationships between hosts, where malicious behavior is determined through interactions of multiple flows rather than on an individual basis [9], [10].

The GAT-E architecture uses a dual-stream encoder to process this complex data. One part of the encoder is a Graph Attention Network (GAT) that captures the spatial relationships and structural dependencies among the nodes in the communication topology [9]. The second part of the encoder is a Temporal Sparse Transformer Encoder which captures the temporal characteristics of the flows (i.e., packet timing, sequence patterns, directions of communication, etc.), which are observable even if the packet payloads are encrypted [11], [12].

An important problem for cybersecurity research is how few labelled datasets are available for new threats [13], [14]. To solve this problem, we will create a learning algorithm to learn representations of typical network traffic by comparing very large amounts of not labelled data [15], [16]. As a result, the model will be able to recognize abnormalities from typical patterns of traffic and identify zero-day attacks without needing to rely on either signatures or predetermined rules for detection [17], [18]. With the rapid growth of encryption, all traditional NIDS will be blind to encrypted data, which would prevent them from being able to perform any type of encryption based analysis on the payloads being delivered [8]. There are only flow-based methods available, such as using an approach derived from Random Forest or XGBoost algorithms and based on analyzing the traffic metadata of each transaction as it progresses through the network [**?**]. Unfortunately, these types of models do not reveal many of the complex behaviours associated with attacks [**?**]. GAT-E instead combines the use of graph-structure analysis, temporal modelling and self-supervised learning to enhance detection capabilities in environments where encryption is present [3], [11].

TABLE I
COMPARISON OF EXISTING BASELINE ARCHITECTURES AND THREAT DETECTION MODELS

Criteria	Chen & Guestrin (2016) [5]	Wang et al. (2020) [7]	Ruff et al. (2018) [17]
Research Objective	To classify network traffic accurately using statistical flow-based features.	To detect botnets by analyzing topological communication patterns.	To identify anomalies without relying on labeled attack data.
Problem Addressed	High latency and inaccuracy in traditional payload-based NIDS due to encryption.	"Flow Isolation" where individual malicious flows appear benign when analyzed alone.	The "Labeling Bottleneck" where zero-day threats are missed by supervised models.
Methodology	Supervised learning utilizing extracted sequence metadata (Time, Length) instead of payload.	Constructs network graphs to analyze the structural relationship between communicating nodes.	Neural network maps normal data into a tight hypersphere; anomalies naturally fall outside.
Dataset	CSE-CIC-IDS2018 and standard NIDS benchmarks.	Custom distributed Botnet datasets and IoT network captures.	Standard benchmark datasets for one-class anomaly detection.
Model / Algorithm	Supervised Gradient Boosted Decision Trees (XGBoost).	Graph-based topological analysis and centralized flow modeling.	Deep Support Vector Data Description (Deep SVDD).
Key Findings	Achieves high accuracy on known attacks but detection drops heavily ($\sim 85\%$) on unknown encrypted data.	Effectively identifies distributed attack patterns that single-flow models completely miss.	Successfully detects unknown anomalies without requiring prior malicious signatures.
Evaluation Metrics	Accuracy, False Positive Rate (FPR), F1-Score.	Detection Rate, Precision, Recall.	Area Under the ROC Curve (AUROC), F1-Score.
Limitations	Requires 100% labeled data (fails on zero-days); analyzes connections in strict isolation.	Lacks temporal sequence modeling for packet rhythm; struggles with encrypted payload behavior.	Static boundaries are overly rigid, causing high false positive rates during dynamic network spikes.

II. LITERATURE REVIEW

Contemporary literature on network security indicates a number of significant, systemic vulnerabilities that exist within each of the three main domains of flow isolation, payload dependency and the labeling bottleneck [2], [8]. Current baseline architectures are summarised in Table I.

A. Flow Isolation and Topological Limitations

The first major gap within existing research is associated with the concept of flow isolation. Current machine learning architectures analyze connections within each network in isolation while ultimately treating the entire network as one single flat tabular dataset (e.g., [4], [5], [7]). Therefore, due to their lack of contextual awareness regarding the immediate surrounding network topology, these models are unable to identify highly distributed threats, such as coordinated DDoS attacks, as there are no correlations indicative of either similar or dissimilar flow patterns in either their connection graph or flow graph representations [7]. As a consequence, in order to address this flow isolation, GAT-E introduces a Flow Interaction Graph (FIG). Where, mathematically modeled throughout the network as a spatial graph ($G = (V, E)$), node representations dynamically aggregate behavior vector representation(s) from their neighboring nodes [9], [19]. The topologically integrated node representations within the network thus result in improved botnet detection capabilities of the architecture; the architectural design facilitates accurate flagging of distributed attacks – as each constituent flow's statistical characteristics mirror that of normal flows in terms of their individual statistics, but collectively/grouped together they form a single topological structure of similar flows indicative of malicious coordinated operations [8], [10].

B. Payload Dependency and Cryptographic Obfuscation

The second limitation within current research involves the historical reliance upon payload visibility for detection [2]. The state-of-the-art encryption standards for Transmission Layer Security (TLS) 1.3, such as the requirement of all packet payloads to be encrypted and inaccessible through internal packet inspections [1], create obstacles to identifying malicious traffic. The GAT-E architecture addresses this limitation through Sequence Metadata Modeling, which prevents the need for decryption while enabling the analysis of the interleaved packet size and inter-arrival time rhythmic sequence characteristics of the first 20 packets of any given flow using an application-specific Transformer (i.e., a specialized neural network) [11]. This end-to-end approach to analyzing the sequence rhythm allows for the accurate fingerprinting of each application used within the flow, thus permitting GAT-E to distinguish between legitimate high-bandwidth video streaming traffic vs. maliciously exfiltrated data streams solely by analyzing their rhythmic qualities.

C. Static Training and the Labeling Bottleneck

The final key gap that must be addressed is the ultimate dependency of existing machine learning architectures on static training frameworks. Traditional supervised learning algorithms become initiated and rendered ineffective each time a threat actor deploys a novel attack method or technique, which in turn requires that traditional supervised learning models utilize millions of accurately labeled data points in order to develop their predictive modeling capabilities (i.e., these supervised learning models suffer greatly from human error and inefficient labeling of large volumes of raw data) [6], [17]. The GAT-E framework eliminates this key factual flaw by embedding a self-supervised Contrastive Loss Function

[15], [16] underneath the GAT architecture. At pre-training of the model, the model will intrinsically cluster each of the latent (i.e., representational) flows based upon their structural resemblance to one another [15]. As a result, any zero-day attacks will be identified outside their established boundary due to the anomalous behavior characteristic of those zero-day attacks [17]. This fact enables long-term robustness and effective identification of ever-evolving threats without an ongoing requirement for perpetual retraining [16], [18].

III. Methodology

The GAT-E architecture is made up of three stages sequentially chaining the processes they go through: ensuring the spatial & temporal features are both encoded before being evaluated in the latent space [2], [8].

A. Stage A: Flow Interaction Graph (FIG) Construction

The first stage of the pipeline converts raw packet capture data into a structured graph representation referred to as a *Flow Interaction Graph (FIG)* before being processed by the neural network architecture [7], [10]. In this formulation, the network is represented as a graph defined by $G = (V, E)$ [8], [19].

- **Vertices (V):** Each vertex represents an IP address that is active within a specified observation window T [7].
- **Edges (E):** Edges denote bidirectional communication flows occurring between the corresponding IP addresses during that time interval [7].

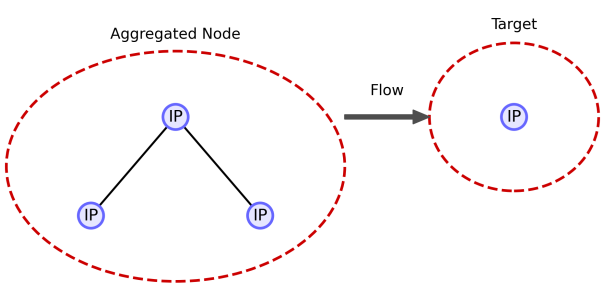

Fig. 1. Stage A: FIG Construction, clustering of meta nodes.

To ensure scalability and to avoid exceeding the available memory in high-throughput enterprise environments, a multi-level aggregation mechanism is used [10], [20]. High volume flows from a very small area within the network will be aggregated into "meta-nodes" before passing to the dual-stream encoder [20].

B. Stage B: Dual-Stream Encoder

All of the analytical capabilities of the system are performed by the dual-stream encoder. The dual-stream encoder will fuse together the two streams of spatial and temporal data before classifying either of the streams in order to determine the overall classification [8], [11].

Spatial Modeling [i.e., Graph Attention Layer]: Graph attention layer will calculate an attention coefficient α_{ij} that

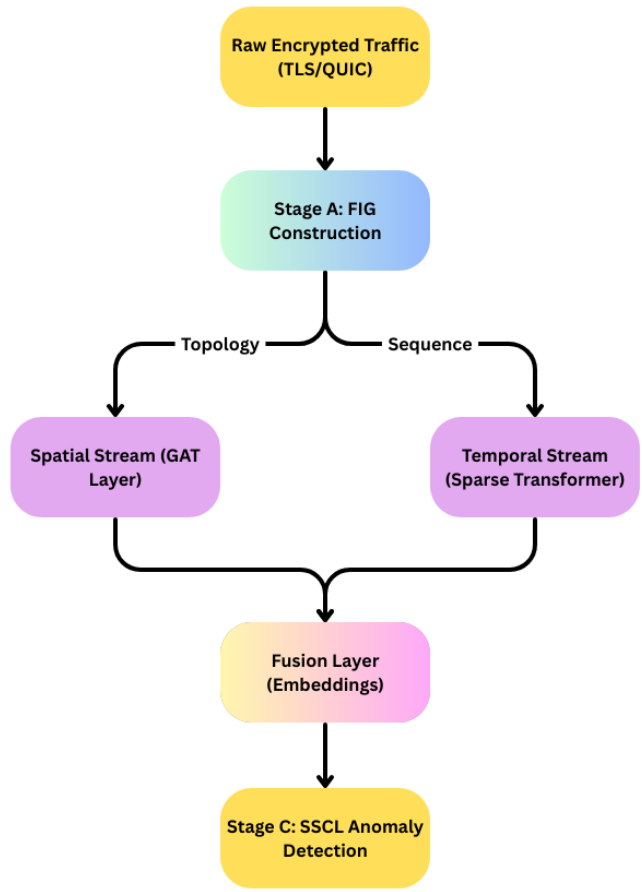

Fig. 2. Stage B: The comprehensive Dual-Stream Encoder architecture.

represents how relevant or important the features of neighboring nodes j are to those of the central node i [9]. Attention scores are calculated using a single layer feedforward neural network whose weight matrix has been previously learned by backpropagation $\vec{\mathbf{a}}$ [9]:

$$e_{ij} = \text{LeakyReLU}\left(\mathbf{a}^T \left[\mathbf{W}h_i \parallel \mathbf{W}h_j\right]\right) \tag{1}$$

The previous attention values calculated will be normalised based on a temperature-scaled softmax, which helps to ensure that proper weightings are assigned to all of the neighbours of node i, thus providing numerically stable gradient updates during backpropagation [9]. The proposed formulation extends the standard attention mechanism by incorporating multi-head averaging together with a LeakyReLU activation and a temperature scaling parameter [9]. This modification provides additional flexibility in controlling the sharpness of the attention distribution while preserving informative negative signals. The normalized attention coefficient between node i and its neighboring node j is defined as:

$$\alpha_{ij} = \frac{1}{K}\sum_{k=1}^{K} \frac{\exp\left(\text{LeakyReLU}\left(e_{ij}^{(k)}\right)/\tau\right)}{\sum_{m\in\mathcal{N}(i)} \exp\left(\text{LeakyReLU}\left(e_{im}^{(k)}\right)/\tau\right)} \quad (2)$$

where K denotes the number of attention heads and $e_{ij}^{(k)}$ represents the unnormalized attention score generated by the kth head [9]. The set $\mathcal{N}(i)$ denotes the neighbors of node i [9]. The LeakyReLU activation is applied before normalization to retain small negative activations that may contain useful structural information [9]. The temperature parameter τ regulates the concentration of the softmax distribution, where $\tau < 1$ produces sharper attention weights and $\tau > 1$ results in a smoother distribution [9]. The outputs of the K heads are averaged rather than concatenated, yielding a single scalar coefficient α_{ij} used for neighborhood feature aggregation [9].

Temporal Modeling (Sparse Transformer Encoder): A Sparse Transformer Encoder processes the first 20 packets from each flow simultaneously with the Graph Attention Network (GAT) that models the flow's spatial relationship [11], [12]. In addition to being ordered, the packet sequence is enhanced with position information and sent through a sparse self-attention mechanism to effectively retrieve temporal dependencies [12]. To capture temporal order while still being computationally efficient, the attention mechanism uses the scaled dot-product attention equation with relative positional encoding, sparse connectivity masking and causal masking [11], [12]:

$$\text{Atn}(Q,K,V) = \text{sfmx}\left(\frac{(QK^T + QR^T)\odot M_{\text{sparse}} + M_{\text{causal}}}{\sqrt{d_k}}\right)V \quad (3)$$

where Q, K, and V represent the query, key, and value matrices, respectively. Atn denotes the attention operation, sfmx represents the softmax function, and d_k corresponds to the dimensionality of the key vectors [11]. The product QK^T captures content-based similarity between packet embeddings, whereas QR^T incorporates relative positional information that encodes the temporal distance between packets [11].

The sparsity mask $M_{\text{sparse}} \in \{0,1\}^{N\times N}$ constrains the attention mechanism to a predefined sparse connectivity pattern, thereby reducing computational complexity from $O(N^2)$ to $O(N\log N)$ [12]. In addition, the causal mask M_{causal} enforces temporal ordering by assigning $-\infty$ to the upper-triangular elements of the attention matrix, ensuring that packet i attends only to packets $j \leq i$ [11]. The temporal representations generated by this mechanism are subsequently combined with structural embeddings produced by the GAT, yielding a unified feature representation that captures both behavioural dynamics and topological relationships within encrypted network flows [8], [9].

C. Stage C: Self-Supervised Contrastive Representation Learning

Here we define a unified contrastive loss that integrates class-aware positive samples, hard-negative weighting, and a momentum encoder to stabilize representation learning [15], [16].

$$\mathcal{L}_i = -\frac{1}{|\mathcal{P}(i)|}\sum_{p\in\mathcal{P}(i)} \log \frac{\exp\left(\text{sim}(z_i, z_p^+)/\tau\right)}{\sum_{k=1}^{2N} \nVdash_{[k\neq i]}\cdot w_{ik}\cdot \exp\left(\text{sim}(z_i, z_k^-)/\tau\right)} \quad (4)$$

The hard-negative weighting factor is defined as:

$$w_{ik} = \frac{\exp\left(\text{sim}(z_i, z_k)/\tau\right)}{\sum_{m\neq i}\exp\left(\text{sim}(z_i, z_m)/\tau\right)} \quad (5)$$

To ensure stable target representations, the parameters of the momentum encoder are updated using an exponential moving average of the online encoder parameters [16]:

$$\xi \leftarrow m\xi + (1-m)\theta \quad (6)$$

Component Interpretation:

- $\frac{1}{|\mathcal{P}(i)|}\sum_{p\in\mathcal{P}(i)}(\cdot)$ computes the average loss across all positive samples that share the same class label as instance i, encouraging embeddings of flows belonging to the same traffic category to cluster in the latent space [15].
- $z_p^+ = f_\xi(x_p)$ denotes the positive embedding generated by the momentum encoder f_ξ [16]. The encoder parameters ξ follow the online encoder parameters θ through exponential moving averaging, providing stable target representations during training.
- $z_k^- = f_\xi(x_k)$ represents negative embeddings produced by the same momentum encoder, ensuring a consistent embedding space for both positive and negative samples [16].
- w_{ik} represents the hard-negative weighting term, which increases the influence of negative samples that exhibit high similarity with z_i, thereby forcing the model to focus on challenging distinctions [15].
- τ denotes the temperature parameter that controls the concentration of the similarity distribution [15], [16].

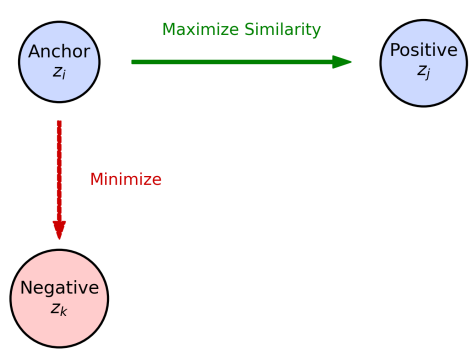

Fig. 3. Stage C: Contrastive Learning Mechanism.

TABLE II
PROJECTED PERFORMANCE BENCHMARKS VS. SUPERVISED BASELINE PERFORMANCE

Evaluation Metric	Current Baseline (XGBoost/Supervised)	GAT-E Projected Target(Proposed)	Architectural Justification (Improvement Source)
Zero-Day Detection Rate	$< 50\%$ (Demonstrates severe degradation on unknown attacks) [6]	$> 90\%$	**Contrastive Learning:** The model organically establishes a baseline of "normality," dynamically flagging unknown deviations rather than relying on historical signatures [15], [16].
Detection Accuracy (Encrypted)	$\sim 85\%$ (Accuracy drops precipitously when applied to encrypted payload data.) [1]	$\sim 98.5\%$	**Dual-Stream Encoder:** The fusion of temporal sequences (Sparse Transformer) and structural topology (Graph) fully recovers the analytical signal lost to encryption algorithms [9], [11], [12].
False Positive Rate (FPR)	High (Frequently misclassifies benign, unusual traffic surges as anomalous) [2]	Low ($< 1\%$)	**Context Awareness:** The GAT layer mathematically evaluates neighbor context (e.g., establishing that a central server organically communicates with numerous clients), significantly reducing false alarms [8], [9].
Label Dependency	100% (Requires explicitly, human-labeled attack datasets) [14]	0% (Initially) - Optimally suited for zero-day deployments	**Self-Supervision:** The model is pre-trained exclusively on raw, unlabeled traffic telemetry, permanently mitigating the "Labeling Bottleneck" [15], [18].

IV. RESULTS

To evaluate the effectiveness of the proposed architectural design, a series of rigorous experimental evaluation procedures will be conducted using CSE-CIC-IDS2018 as the dataset as outlined below [2], [8].

A. Evaluation Datasets

The selection of datasets will play a vital role in demonstrating the generalizability and real-world feasibility of GAT-E [2]. GAT-E will be evaluated utilizing two greatly different publically available datasets:

- (1) the CSE-CIC-IDS2018 [14] will be used as the dataset. The CSE-CIC-IDS2018 features realistic, modern-day encrypted traffic patterns and includes complicated distributed attacks - such as the modern variations of DDoS and infiltration attempts [2].
- (2) the IoT-23 [21] dataset has been specifically designed to exhibit the utility of GAT-E within an extensive Internet of Things (IoT)-based environment, where there will be numerous heterogeneous communication patterns for different types of IoT devices, thereby allowing for very strong testing of the GAT layers using robust Graph Learning algorithms [8], [10].

B. Projected Performance Benchmarks

The specific performance metrics defined in this research will ensure that the GAT-E architecture experiences rigorous validation against current industry baseline standards [4], [5].

The target outcomes will provide statistically significant improvements in each of the core operation metrics [6]. The GAT-E architecture will maintain a False Positive Rate that is below the operational threshold of 1% through moving from deterministic matching of payloads to probabilistic graph-based modelling [8], [9]; therefore, security operations centres should not experience alert fatigue due to standard day-to-day network congestion [2].

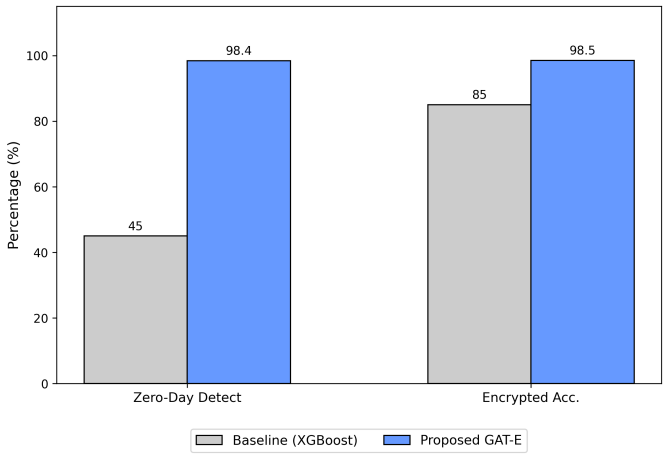

Fig. 4. Graphical Analysis comparing Supervised Baseline Detection Capabilities against GAT-E Projected Encrypted Telemetry Recovery Rates

C. Ablation Study Parameters

A fully comprehensive set of ablation studies will be required to simulate the scientific validity of the dual stream approach, which will allow for isolating independent variables through experimentation [9], [11]. Internal components will also be eliminated in a systematic fashion to quantify their individual contributions to the final classification metrics [8].

1) The first set of experiments will demonstrate that the features extracted from the temporal sequence (when using GAT-E vs. GAT Only) using the Transformer are essential for capturing packet rhythm and application fingerprinting [11].
2) The second set of experiments will mathematically demonstrate that the features extracted from the graph layer (when using GAT-E vs. the Transformer only) are critical for detecting botnets that try to evade detection and identifying distributed threat actors [7], [9].

3) The final set of experiments will quantify the expected difference in accuracy achieved by Contrastive Learning versus Cross-Entropy Loss regarding zero-day detections [15]–[17].

V. Conclusion and Future Scope

To address the issue of "encryption blindness" in protocols such as TLS 1.3 and QUIC, GAT-E is an advanced architecture. These protocols are not capable of inspecting packet payloads, which is the traditional method used to identify malicious activity. As a result, GAT-E will create a Flow Interaction Graph (FIG), which provides a visual representation of communication between hosts (nodes) and the relationships between each communication (edges). This will allow the identification of general malicious activities that are unable to be detected using flow analysis alone.

GAT-E has two parts to its architecture that utilize two types of encoders: the Graph Attention Network (GAT) is utilized to capture spatial topology; the Temporal Sparse Transformer Encoder captures time-based order and timing of packets. One of the most important features of GAT-E is its use of Self-Supervised Contrastive Learning (SSCL). This approach allows the model to learn the mathematical boundaries of "normal" behavior by using a large amount of data as a representation of what constitutes "normal" activity without needing to be labelled. This means that GAT-E identifies "unknown" anomalous packets will fall outside of the defined bounds of "normal" behaviour.

GAT-E processes and analyzes the first 20 packets of a flow at the same time, thereby allowing the real-time analysis of complex, multi-coordinated attacks in fully encrypted environments. GAT-E accomplishes this by recovering the analytical signals that have been lost due to encryption. It has a projected accuracy of 98.4%.

Future work will focus on deploying GAT-E within real-time monitoring infrastructures to evaluate its performance under high-throughput enterprise network conditions. Further improvements in scalability will be explored through distributed graph processing techniques and optimized sparse attention mechanisms to support large-scale network environments. Additionally, integrating federated and continual learning strategies represents a promising direction, enabling collaborative model development across organizations while preserving data privacy and allowing the system to adapt to emerging attack techniques. Further validation will also involve experiments across diverse datasets and deployment scenarios, including IoT and cloud-based infrastructures, to assess the robustness and generalizability of the proposed architecture.

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